

Paridarshan Sahoo

Bachelor of Technology
Computer Science and Engineering - Data Science
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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate and Smart India Hackathon 2023 Finalist with hands-on experience building scalable REST APIs and real-time backend systems using Java and Spring Boot. Proficient in data structures, algorithms, and software engineering principles. Passionate about writing clean, maintainable, and production-ready backend software.

EDUCATION

• **Bachelor of Technology - Computer Science and Engineering - Data Science** CGPA: 7.0/10
Maharana Pratap Engineering College, Kanpur 2022–2026

TECHNICAL SKILLS

Languages: Python, Java, SQL
Backend: Spring Boot, REST APIs, WebSocket, Microservices
Databases: PostgreSQL, PostGIS, Redis
Tools: Git, GitHub, Docker, Kafka, Linux (Debian), Postman
Concepts: Data Structures & Algorithms, OOP, Distributed Systems, JWT

PROJECTS

• Payment Processing System 🔄

Technologies Used: Java, Spring Boot, PostgreSQL, Redis, Apache Kafka, JWT, Docker

- Architected a microservices-based payment platform with 3 independent services — Account Service, Transaction Service, and Notification Service — each with its own dedicated PostgreSQL database, JWT-secured REST APIs, and containerized via Docker Compose.
- Implemented Redis distributed locking using SETNX to prevent double spending across concurrent requests, and a multi-layer idempotency system (Redis cache as fast path, PostgreSQL as fallback) ensuring duplicate payment requests never result in double charges; published payment events to Kafka for decoupled, event-driven email notifications.

• Real-Time Public Transport ETA & Smart Ticketing System 🔄

Technologies Used: Java, Spring Boot, PostgreSQL, PostGIS, Redis, WebSocket, JWT, Docker

- Designed and built 15+ REST APIs for real-time bus tracking, ETA prediction, and crowd density estimation, with JWT-based RBAC across three user roles — PASSENGER, DRIVER, and ADMIN — secured via Spring Security.
- Built a WebSocket broadcast system for live bus location and occupancy updates, and an offline-first ticketing sync API for low-connectivity scenarios with admin analytics endpoints covering revenue and route performance.

• TorNet Granted - Enumeration of Hidden Servers 🔄

Technologies Used: Python, Flask, BeautifulSoup, Tor, SQLite

- Developed a recursive graph traversal engine using Python and Tor proxy to navigate .onion networks, targeting deep-web nodes not indexed by public directories, with a differential validation engine to isolate and health-check unlisted active servers.
- Built a DOM extraction module using BeautifulSoup to scrape and store metadata from verified hidden services in SQLite for security analysis.

CERTIFICATIONS

- Python For Data Science from NPTEL by IIT Madras
- Data Structures and Algorithms using Java from NPTEL by IIT Kharagpur

ACHIEVEMENTS

- **Finalist, Smart India Hackathon (SIH) 2023:** Selected for the Grand Finale competing against 400+ teams; built a solution for enumerating hidden servers on Tor-based networks.
- **Ranked Top 5, Code-o-Fiesta 2.0:** Top 5 out of 300+ participants in a competitive coding contest organized by MPGI, Kanpur.
- **Hackathon Finalist:** Built working prototypes at HackIndia24 (Spark 7) and Ace Hack 4.0 under strict time constraints at national level.